Science

Teaching and Learning Syllabus

Grades 3 to 6

Mauritius Institute of Education
under the aegis of
Ministry of Education and Human Resources, Tertiary Education and Scientific Research

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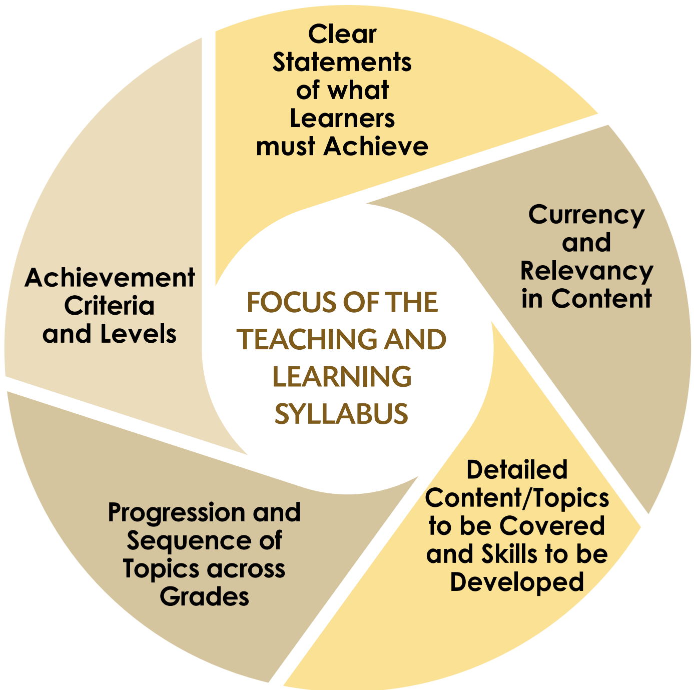
Science

The Teaching and Learning Syllabus refers to the content of what is to be taught, and to the knowledge, skills, competencies, attitudes and values that are to be developed. It articulates concisely the specific learning outcomes for each grade as per the NCF 2015 and features achievable learning outcomes that represent a Teaching and Learning road map.

• The focus of the TLS



• Clear statements of what learners must achieve

The TLS spells out the specific learning outcomes that learners are expected to achieve progressively in terms of knowledge, understanding, skills, 21st Century Competencies, attitudes and values during each year of schooling, from Grades 1 to 6.

• Currency and relevancy in content

The TLS ensures currency and relevance of content and acquisition of developmentally-appropriate concepts and skills.

• Detailed content/topics to be covered and skills to be developed

The TLS details the content and topics to be covered as well as the skills to be developed for each grade while addressing the problem of curriculum overload and stress.

- **Progression and sequence of topics across grades**

The TLS outlines the scope and sequence of the specific learning outcomes for each grade. It indicates the grade in which knowledge, concepts, skills and processes related to the learning outcomes are introduced, based on a spiral curriculum. The learning outcomes are progressively sequenced across grades.

- **Achievement criteria and levels**

The TLS defines assessment criteria that are scaffolded and represent clear measurable outcomes. The levels reflect the competencies in terms of knowledge, skills, values and attitudes expected to have been acquired by learners. The development of achievement levels are meant to assist in the differentiation of teaching, learning and assessment so that the curriculum is accessible to all learners.

Contents

RATIONALE

The syllabus represents a teaching and learning road map for science for Grades 3 to 6. It focusses on the learning outcomes as stated in the *National Curriculum Framework - Grades 1 to 6.* It refers to the knowledge, inquiry skills, attitudes and values which are to be developed in children and provides clear statements of what they must **know** and be **able to do**. The scope, progression and sequence of content and skills across grades are detailed in this syllabus. The development of achievement levels is meant to assist in the **differentiation of teaching, learning and assessment** so that the science curriculum is accessible to **all** learners.

THE DEVELOPMENT OF 21ST CENTURY COMPETENCIES AND THE SCIENCE CURRICULUM

The National Curriculum Framework for Nine-Year Continuous Basic Education (2015) identifies several competencies - apart from those associated with different learning areas - that all children and young people should develop. In the science curriculum, current and relevant content for the 21st century learner has been carefully chosen to help in developing 21st Century Competencies. Learners will be given the opportunity to explore science concepts and engage in scientific activities and in the process develop these competencies. Content, pedagogy and assessment are all aligned to the task of developing these competencies.

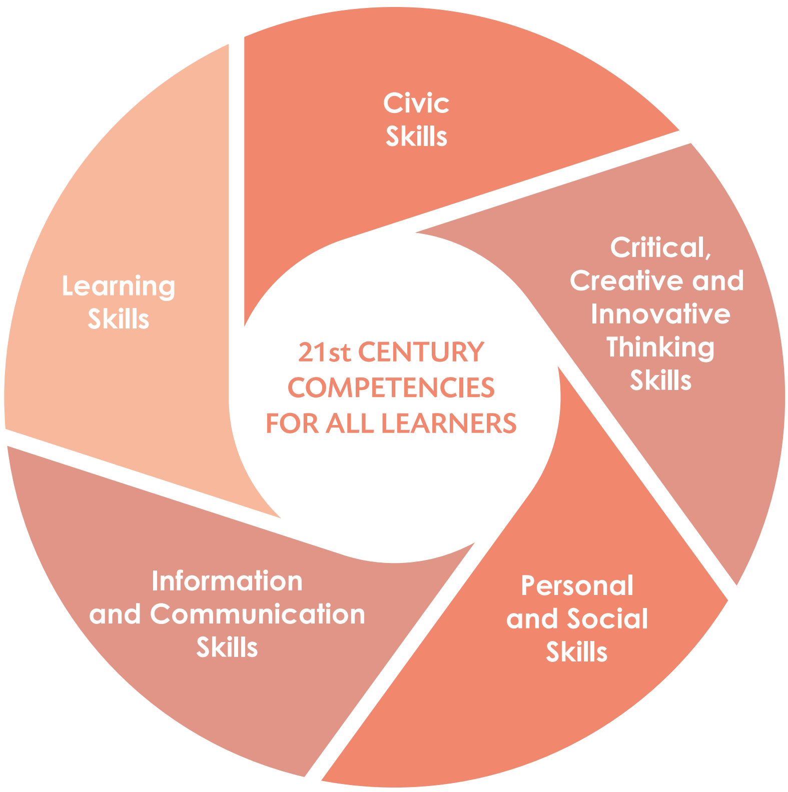


Figure 1: 21st Century Competencies for All Learners

EXPECTED LEARNING OUTCOMES FOR SCIENCE

The expected learning outcomes for science are spelt out below. Table 1 represents the scope and sequence of the expected learning outcomes across the different grades.

EXPECTED LEARNING OUTCOMES	GRADES			
	3	4	5	6
Demonstrate knowledge and understanding of scientific concepts.	•	•	•	•
Show an understanding of the application of basic scientific concepts in everyday situations.		•	•	•
Show an understanding of some observable phenomena in the natural physical environment.	•	•	•	•
Show an understanding of the interdependence of living things and their environment.			•	•
Demonstrate an understanding of the importance of safety in scientific activities.			•	•
Demonstrate an appreciation of the importance of conserving habitat and environment.		•	•	•
Use ICT tools to search for, collect and analyse relevant scientific information and ideas.	•	•	•	•
Develop and use a range of inquiry skills.	•	•	•	•
Care for planet Earth.	•	•	•	•

Table 1: Expected Learning Outcomes at the end of Grade 6

SCOPE AND SEQUENCE OF CONTENT AREAS FOR SCIENCE

The science syllabus is organised in three main strands:

- Knowledge and Understanding
- Inquiry Skills and Processes
- Attitudes and Values

KNOWLEDGE AND UNDERSTANDING

Some fundamental topics in science provide an edifice upon which all scientific concepts and knowledge can be constructed. In addition, these topics help learners make connections to concepts learned in other areas such as mathematics and geography. Using a spiral approach, students will deepen their understanding of these fundamental topics as they progress through the science curriculum from Grades 3 to 6.
The key topics that are addressed in the curriculum for science are:

- The Senses
- Air
- Water
- Living and Non-Living Things
- Plants
- Animals
- Materials in our Environment
- Energy
- The Simple Electric Circuit
- Ecosystems of Forests and Lagoons
- Protection and Conservation of the Environment
- The Earth, Moon and Sun in our Solar System

- Inquiry Skills and Processes

Questioning Asking questions about animals, plants, objects and events in the immediate environment which may lead to investigations.	**Predicting** Suggesting outcomes of an investigation, based on observations and prior knowledge.
Observing Using the senses to gather information from the immediate environment.	**Investigating and experimenting** Carrying out simple investigations.
Classifying Grouping objects according to common observable characteristics or hierarchical relationships.	**Measuring** Selecting and handling simple apparatus for various tasks.
Comparing Identifying similarities and differences between objects,concepts or processes.	**Drawing conclusions** Explaining observations or data and drawing conclusions from simple investigations.
Communicating Receiving and transmitting information orally, in writing or by drawing pictures, tables and models.	**Explaining based on observations** Giving a general explanation for what is observed.
Problem-Solving Considering a problem and finding a solution.	**Decision-making** Analysing different options and making a choice.

Table 2: Description of Inquiry Skills and Processes

- Attitudes and Values

Curiosity Asking questions about objects and events and finding out more about events and objects on their own.	**Respect For Evidence** Explaining their results and conclusions and listening to other students' results and explanations.
Persistence Completing activities and persisting at tasks.	**Respect for Living Things and the Environment** Showing sensitivity to living things and responsibility towards the environment.
Cooperation Sharing with others and working together with others.	**Concern For Safety** Observing safety instructions.

Table 3: Scientific Attitudes and Values

Table 4 indicates the grades at which fundamental topics, skills and processes, attitudes and values are introduced, sequenced and progressively developed.

Content Areas	Grade Level			
	Grade 3	Grade 4	Grade 5	Grade 6
Knowledge and Understanding				
The Senses	•			
Air	•	•		•
Water	•	•	•	
Living and Non-Living Things	•	•		
Plants and Animals	•	•	•	•
Materials in our Environment	•	•		•
Energy		•	•	•
The Simple Electric Circuit			•	
Ecosystems of Forests and Lagoons				•
Protection and Conservation of the Environment	•		•	•
The Earth, Moon and Sun in our Solar System				•
Inquiry Skills and Processes				
Questioning	•	•	•	•
Observing	•	•	•	•
Classifying	•	•	•	•
Comparing	•	•	•	•
Communicating	•	•	•	•
Predicting			•	•
Investigating and Experimenting	•	•	•	•
Measuring		•	•	•
Drawing Conclusions		•	•	•
Explaining Based on Observations		•	•	•
Decision-making		•	•	•
Problem-Solving	•	•	•	•
Attitudes and Values				
Curiosity	•	•	•	•
Persistence			•	•
Cooperation	•	•	•	•
Respect for Evidence			•	•
Respect for Living Things	•	•	•	•
Respect for the Environment	•	•	•	•
Concern for Safety	•	•	•	•

Table 4: Progressive Sequencing of Fundamental Topics, Inquiry Skills, Processes, Attitudes and Values

ACHIEVEMENT LEVELS

Achievement Levels 1, 2 and 3 are scaffolded learning objectives that represent clear measurable learning outcomes. The levels reflect the competencies in terms of knowledge, inquiry skills, values and attitudes expected of pupils at the end of each grade. The criteria move progressively from Level 1 to Level 3.

FUNDAMENTAL TOPICS, SUBTOPICS, SPECIFIC LEARNING OUTCOMES AND ACHIEVEMENT LEVELS FOR GRADE 3

Fundamental Topics	Subtopics	Specific Learning Outcomes	Achievement Level 1	Achievement Level 2	Achievement Level 3
The Senses	The five sense organs The five senses	Identify and name the main sense organs. Relate the senses to their functions.	Name the different parts of the body that help us observe the environment. Identify, name and draw the five sense organs. Identify and name the five senses. Show understanding of the use of the sense organs.	State the function of each sense organ. Relate the five senses to how we see, hear, touch, smell and taste. Investigate how we use the senses and communicate findings. Show understanding of the use of the five senses.	Show understanding of how we use the eyes, ears, nose, tongue and skin to learn about and gather information from the environment around us. Compare the five senses.
Living and Non-Living Things	Characteristics of livings things (grow, need food and breathe) Classification of living things into plants and animals	Recognise and identify a variety of things in the immediate environment. Observe and classify living and non-living things and state differences between them.	State that there is a large variety of things in our environment. Recognise and name living and non-living things. Observe and identify living and non-living things. State that there is a large variety of plants and animals around us. Give examples of animals and plants.	Compare and classify living and non-living things. Compare and classify living things into animals and plants. Show understanding that living things grow, need food and breathe. State the uses of some non-living things.	Infer that living things grow, need food and breathe. Compare living and non-living things. Demonstrate skills to care for animals and plants in the environment.

Materials in our Environment	Wood, plastic, paper and glass	Observe and name a variety of common materials. Identify the types of common materials.	Observe and identify wood, plastic, paper and glass in everyday life.	State the uses of wood, plastic, paper and glass.	Compare and classify objects made of wood, plastic, paper and glass.
Water	Physical properties of water Sources of water Uses of water Water conservation	Investigate and recognise the presence of water around us. Demonstrate understanding of the need to conserve water.	Identify and name some common sources of water. List some common uses of water. State the physical properties of water.	State how water is obtained from rain. Investigate the presence of water around us and communicate findings.	Compare sea and fresh water. State why we must not waste water.
Air	Presence of air	Investigate and recognise the presence of air around us.	State that air is all around us. State that we breathe air all the time. List some common uses of air.	Demonstrate understanding of some common uses of air. State that air is present in empty bottles, soil and water. Investigate the presence of air around us and communicate findings.	Infer the presence of air through observations and simple experiments.
Protection and Conservation of the Environment	Problems in the environment Protecting the environment	Recognise ways to keep our environment clean.	Explore and observe a given environment. List some problems observed in the environment such as domestic wastes, noise, smoke, dirty and stagnant water.	Explain how air and water can become dirty. Suggest ways to solve the problem of dirty air and water. Infer that breathing dirty air is bad for health. Infer that dirty water can affect our health. List ways to keep our environment clean.	Infer how the problems in the environment affect people and other living things. Show concern for keeping our environment clean.

FUNDAMENTAL TOPICS, SUBTOPICS, SPECIFIC LEARNING OUTCOMES AND ACHIEVEMENT LEVELS FOR GRADE 4

Fundamental Topics	Subtopics	Specific Learning Outcomes	Achievement Level 1	Achievement Level 2	Achievement Level 3
Living Things	Characteristics of livings things Classification of living things	Describe a few characteristics of living things.	State futher characteristics of living things (move, reproduce).	Investigate living things and communicate findings. Compare and classify living things.	Infer the differences between plants and animals.
Animals	Why and how animals move Endemic and exotic animals Protection of exotic and endemic animals	Explain the need for animals to move. State the different ways animals move. Show understanding of exotic and endemic animals. List measures taken to protect our exotic and endemic animals.	Observe animals and look for body features that help them to move. Recognise and name some endemic and exotic animals in Mauritius and Rodrigues. Define the terms exotic and endemic.	State reasons for animals to move. List and name the features which help animals to move around. Show understanding of the terms exotic and endemic. List measures to protect endemic and exotic animals.	Observe and relate certain body features with ways animals move. Compare and classify animals as endemic and exotic. Show understanding of the need to protect animals.
Plants	Parts of a flowering plant The functions of the flower, fruit and seed Exotic and endemic plants Protection of endemic plants	Observe and investigate the different parts of a plant and communicate findings. Show understanding of exotic and endemic plants. List measures taken to protect our exotic and endemic plants.	State some uses of plants. Observe the different parts of a flowering plant. Draw a plant to show the root, the stem, the leaves, the flower and the fruit.	State the functions of the flower, fruit and seed. Investigate the functions of the flower, fruit and seed and communicate findings. Recognise and name endemic and exotic plants. List measures to protect endemic and exotic plants.	Infer the importance of protecting our endemic plants. Infer how plants are useful to people and animals. Show understanding of the need to protect our endemic plants. Compare and classify plants as endemic and exotic.
Water	Importance of water Three states of water - ice, liquid water and water vapour Boiling of water Filtration and purification of water Common solids dissolving in water	Demonstrate an understanding of the importance of water. Compare the three states of water.	Recall the importance of water in our daily activities. Recognise and name the three states of water. Identify different types of water pollution.	Investigate and infer the properties of pure water and communicate findings. Investigate the three states of water and communicate findings. Compare the three states of water. Demonstrate understanding of filtration and purification of water.	Infer the importance of water. Demonstrate understanding of the need to filtrate and purify water. Investigate the boiling of water and communicate findings. Investigate the dissolving of some common substances in water and communicate findings.

Air	Importance of air Burning	Demonstrate an understanding of the importance of air.	Recall that air is present everywhere. State that air is colourless, odourless and tasteless.	Investigate that air is important for living things.	Infer the importance of air for plants and animals. Infer from experiments that air is important for burning.
Materials in our environment	Physical properties of materials Choosing the right materials to make objects	Compare the physical properties of a variety of everyday materials. Choose the right materials to make things.	Observe and identify different materials in everyday life (wood, paper, glass, plastic, metals, rubber).	Investigate the properties of some materials. Infer that different materials have different properties. Relate the properties of different materials to their uses.	Demonstrate an understanding of how the use of a material depends on its properties. Select the right materials to make objects.
Energy	Sources of energy Forms of energy	Recognise, list and identify different sources of energy. Recognise, list and identify different forms of energy.	State some uses of energy. Name different sources of energy (sun, wind, petrol). Name different forms of energy (light, heat and movement).	Investigate different sources of energy. Investigate different forms of energy.	Infer that energy is useful to all living things.
Protection and conservation of the environment	Human impact on our environment Environmental pollution	Recognise ways in which human activities affect the environment.	Explore and observe an area near the school or place where you live. List the types of environmental pollution. List causes of different types of pollution.	Infer why these types of pollution exist. Relate human activities to different types of pollution. Demonstrate understanding of the different types of pollution.	State measures to have a pollution-free environment. Show concern for the protection of the environment. Infer effects of pollution on living things and the environment.

FUNDAMENTAL TOPICS, SUBTOPICS, SPECIFIC LEARNING OUTCOMES AND ACHIEVEMENT LEVELS FOR GRADE 5

Fundamental Topics	Subtopics	Specific Learning Outcomes	Achievement Level 1	Achievement Level 2	Achievement Level 3
Animals	Habitats of animals Endangered and rare animals in Mauritius and Rodrigues Measures for protecting endangered animals	Demonstrate appreciation that most animals live in habitats that are suitable for them. List measures taken to protect our endangered animals.	Explore and observe animals which live in different habitats and communicate findings. List the different habitats of animals. Name some endangered animals of Mauritius and Rodrigues. Appreciate the importance of habitats to animals.	Demonstrate understanding of the need to protect endangered and rare animals. Demonstrate understanding of the need to protect, preserve and restore habitats of animals. State measures to protect endangered animals.	Investigate the features of animals which enable them to live in a particular habitat. Infer what happens if the habitat of animals is changed. Infer that animals live in suitable habitats.
Plants	Flowering and non-flowering plants in our environment Parts of a plant Functions of the root and stem of a plant Germination Conditions for plants to grow Soil erosion	Demonstrate appreciation that most plants live in habitats that are suitable for them. Identify and state the functions of the main parts of a plant. Investigate and recognise the conditions for plants to grow and communicate results.	Identify and name the parts of a plant. Describe the parts of a flower. State the functions of the parts of a plant. List the conditions necessary for plants to germinate. List the conditions necessary for plants to grow well. State what is soil erosion. State the causes of soil erosion.	Investigate that one of the main function of the root and stem is to transport water and minerals. Observe and describe how a seed grows into a plant. Draw a simple diagram to show the life cycle of a plant. Identify soil erosion. List measures to prevent soil erosion in Mauritius and Rodrigues. Relate human activities to soil erosion.	Investigate the conditions necessary for plants to germinate and communicate findings. Investigate the conditions necessary for plants to grow well and communicate findings. Compare and classify plants by looking at their different parts. Demonstrate understanding of the effects of soil erosion on living things and the environment.

Energy	Sources of energy Energy from the Sun Fossil fuels Forms of energy Transformation of of energy Conserving energy	Recognise the importance of the Sun and fossil fuels as sources of energy. Identify and list different forms of energy. Demonstrate understanding that energy can be transformed from one form to another. State some measures that we can take to save energy.	Name and list sources and forms of energy. Define the different forms of energy. State that the energy from the Sun is widely used for our daily energy needs. State that fossil fuels are widely used. List measures that we can take to save energy.	Name the forms of energy obtained from batteries, food, Sun, wood, wind, falling water and fossil fuels. Differentiate between sources of energy and forms of energy.	Explain energy transformation in common devices. Investigate the different forms of energy and communicate findings. Infer the differences between forms and sources of energy.
The Simple Electric Circuit	The complete electric circuit Components of a circuit Closed and open circuits Insulators and conductors Safe use of electricity	Identify electric components and use them to light a bulb. Identify some materials that allow a bulb to light.	Identify, list and draw components that can make a complete circuit. Identify and name materials that allow a bulb to light in a complete circuit. Describe the safe use of electricity at home and at school.	Construct and draw a complete circuit to light a bulb. Draw and label the diagram of a complete circuit.	Investigate the simple electric circuit and communicate findings. Differentiate between open and closed circuits. Demonstrate understanding of how electricity flows in a complete circuit. Differentiate between insulators and conductors.
Water	Melting, evaporation, condensation and freezing The water cycle Conserving water	Recognise the interchangeable states of water. Recognise the importance of the water cycle.	Recall the three states of water. Define melting evaporation, condensation and freezing. State some measures that we can take to save water.	Infer that all states of water are interchangeable. Show an understanding of melting, freezing, evaporation and condensation as changes of states. Demonstrate understanding of the importance of saving water.	Explore how salt is obtained from sea water by evaporation. Demonstrate an understanding of the different steps of the water cycle. Explain the importance of the water cycle in maintaining the amount of water on earth.

FUNDAMENTAL TOPICS, SUBTOPICS, SPECIFIC LEARNING OUTCOMES AND ACHIEVEMENT LEVELS FOR GRADE 6

Fundamental Topics	Subtopics	Specific Learning Outcomes	Achievement Level 1	Achievement Level 2	Achievement Level 3
Air	Importance of air Air pressure Extinguishing fires Air Pollution	State that air is a mixture of gases.	Recall the importance of air. State that air is a mixture of several gases. State some gases present in air: nitrogen, oxygen, carbon dioxide and water vapour. Identify and list different causes of air pollution. List different types of air pollution.	Investigate the ways in which we can extinguish different types of fires. Investigate air pressure and communicate findings. Investigate measures to prevent air pollution. Relate human activities to air pollution. Demonstrate understanding of the different types of air pollution.	Demonstrate understanding that oxygen is essential for all living things. Infer some uses of air pressure. Demonstrate understanding of the effects of air pollution. Infer the effects of air pollution on living things and the environment.
Materials in our Environment	Natural and man-made materials Useful properties of natural and man-made materials Rusting of iron Waste disposal	Distinguish between natural and man-made materials. Relate the properties of materials to their uses.	Identify and name some natural and man-made materials. List uses of common materials. State the sources of natural materials. Define rusting. Identify and list different types of environmental wastes.	Recognise that some materials are obtained naturally while others are man-made. Compare natural and man-made materials. Show understanding of how rusting can be prevented. Demonstrate understanding how environmental wastes affect planet Earth.	Investigate the rusting of iron and communicate findings. Relate the properties of materials to their common uses. Infer the use of appropriate materials to make objects Demonstrate understanding of the need to care for planet Earth.
Animals	Characteristics of animals Animals and their food Food groups The human teeth	Differentiate the ways in which animals obtain their food.	Identify different types of animals. List ways in which animals obtain their food. Identify and name the food groups. Identify the milk and permanent teeth in humans.	Classify animals to the type of food they eat. Classify food into different food groups. Demonstrate understanding of a balanced diet. Describe the different types of human teeth Classify animals in different groups. Demonstrate understanding of the functions and importance of the human teeth.	Infer how animals help to maintain the composition of the air. Infer how humans use animals. Infer the importance of a balanced diet. Compare the milk and permanent teeth.

Plants	Functions of the leaf Photosynthesis Composition of the air Breathing in plants	Differentiate the ways in which plants obtain their food.	State the functions of the leaf. State how plants make their food. Define photosynthesis. State the conditions for plants to make their food.	Investigate photosynthesis. Demonstrate understanding of how plants make their food by photosynthesis. Demonstrate understanding of how plants breathe.	Infer how plants help to maintain the composition of the air. Infer the uses of plants.
Energy	Transformation of energy Polluting and non-polluting sources of energy Renewable and non-renewable sources of energy Green energy	Compare hydro-electric and thermal power stations. Recognise polluting and non-polluting sources of energy. Recognise renewable and non-renewable sources of energy.	Define a fuel. List the fuels that are used in the production of electricity. Identify sources of energy that cause and do not cause pollution. Define renewable and non-renewable sources. List renewable and non-renewable sources of energy.	Investigate the transformation of energy in a hydro-electric and thermal power station. Demonstrate understanding of renewable and non-renewable energy. Classify sources of energy into renewable and non-renewable. Demonstrate understanding of green energy.	Differentiate between renewable and non-renewable energy. Demonstrate understanding of the importance of green energy.
The Earth, Moon and Sun in the Solar System	The Sun, Earth and Moon Occurrence of day and night Movement of the Earth and the Moon around the Sun Global warming Carbon-dioxide and global warming.	Show simple understanding of the Sun, Earth and Moon system. Recognise the regular movements of Earth (rotating and spinning) and Moon. Explain how Earth's movement causes day and night.	Recognise that Earth is a planet and the Sun is a star. Identify the Sun, the Earth and the Moon in the Solar system. Recognise the regular movements of Earth. State what is global warming. List measures to prevent global warming.	Demonstrate understanding that the Earth rotates around the Sun. Demonstrate understanding that the Earth spins on its axis. Relate human activities to global warming. Relate a rise in the level of carbon dioxide to global warming. Demonstrate understanding of the Sun, Earth and Moon as different celestial objects.	Explain how Earth's movement causes day and night. Relate the movement of the Sun and Earth to the meaning of the day and the year. Demonstrate understanding of the effects of global warming on living things and the environment. Demonstrate understanding of the need to care for planet Earth.

Ecosystems of Forests and Lagoons	Types of Ecosystems Aquatic and terrestrial ecosystems The forest as an ecosystem The lagoon as an ecosystem Importance of forests and lagoons as ecosystems Threat to our forests and lagoons Beach erosion	Show an appreciation of our forests and lagoons as important ecosystems.	Describe an ecosystem using a forest and a lagoon. Identify a terrestrial and a marine ecosystem. State ways in which the forest and lagoon are threatened by human activities. List measures to protect forest and lagoon ecosystems. State what is beach erosion.	Recognise the forest and the lagoon as two important ecosystems. Compare the forest and lagoon as two different ecosystems. Identify beach erosion. List measures to prevent beach erosion in Mauritius and Rodrigues.	Demonstrate understanding of the need to protect and conserve forests and lagoons. Investigate the forest and lagoon ecosystems. Observe and compare living things in different ecosystems. Relate human activities to soil and beach erosion. Demonstrate understanding of the effects of soil and beach erosion on living things and the environment.

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